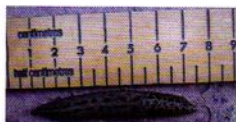


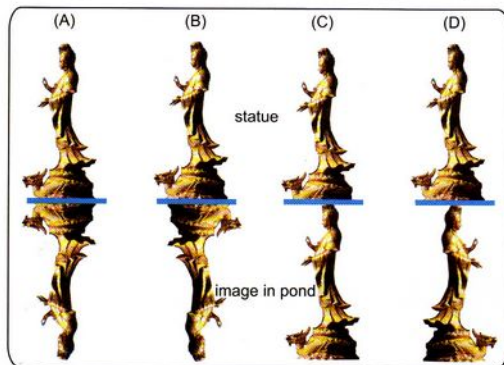
1. What is the approximate length of this animal?



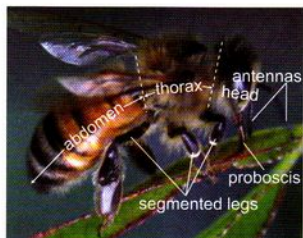
- (A) 7.5 cm
(B) 7.0 cm
(C) 6.5 cm
(D) 6.0 cm

2. Nick observed a statue standing in the middle of a pond and its reflection on the water.

Which diagram shows the statue and its reflected image correctly?



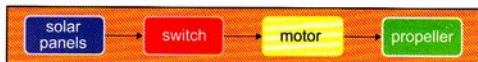
3. The labelled photograph shows some body parts of a honey bee.



To which body part are the segmented legs attached?

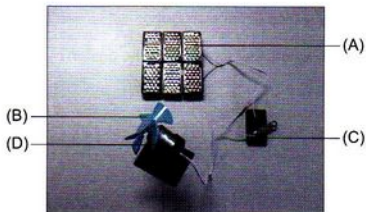
- (A) abdomen
(B) head
(C) proboscis
(D) thorax

4. Tanya drew a chart showing how the parts of her solar-powered project were joined.



The photo shows the parts of her project.

Which part is the switch?



5. Andy wanted to find the volume of a cube. His teacher told him that he could find this by placing the cube in water and measuring how much water it displaced. Volume can be measured in cubic centimetres (cm^3). He set up the experiment below.

block — 



step 1



step 2

What is the volume of the block, in cubic centimetres?

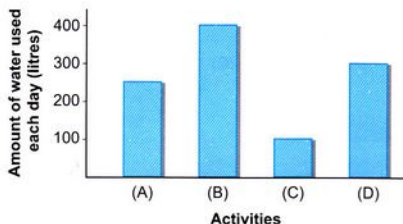
- (A) 1
(B) 2
(C) 6
(D) 8

6. The table shows the amount of water used each day in Kerry's home for different activities.

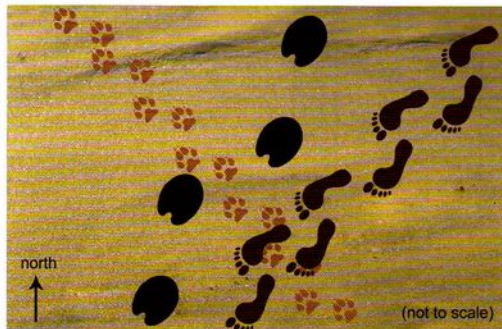
Activities	Amount of water used each day (litres)
washing dishes	100
flushing toilet	250
watering garden	300
taking showers	400

The graph shows the same data.

Which column shows the amount of water used for watering the garden?



7. The diagram shows tracks on a beach.



In which direction did the fox move?

- (A) southeast (B) southwest (C) northeast (D) northwest

8. A group of students made a tally of the things they found in a sandpit.

leaves	
marbles	
rocks	
sticks	

Which option presents the correct data in the best way?

(A)

Number found	Things
leaves	12
marbles	3
rocks	16
sticks	8

(B)

Things	Number found
leaves	16
marbles	3
rocks	12
sticks	8

(C)

Number found	Things
12	leaves
3	marbles
16	rocks
8	sticks

(D)

Things	Number found
leaves	12
marbles	3
rocks	16
sticks	8

9. Matter can exist in three states; solid, liquid or gas, depending on temperature.

The diagram shows how matter can change from one state to another.



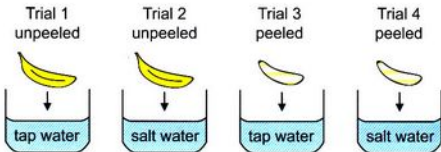
Which labels should be placed in the diagram?

	P	Q	R	S
(A)	gas	solid	heat	cool
(B)	gas	solid	cool	heat
(C)	solid	gas	heat	cool
(D)	solid	gas	cool	heat

10. Mal wants to find the volume of a banana peel.

Mal dropped similar sized unpeeled and peeled bananas into containers of tap water and salt water.

He observed that unpeeled bananas floated just below the surface in tap water, but floated higher in salt water. The peeled bananas sank in both the tap water and the salt water.

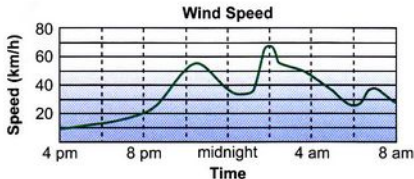


To work out the volume of a banana peel, he used the changes in water levels in the containers.

Which trials were the best ones for Mal to use?

- (A) Trials 1 and 2
(B) Trials 1 and 3
(C) Trials 2 and 3
(D) Trials 2 and 4

11. The wind speed was recorded near a town that was 1970 m above sea level.



The table shows the types of winds of different speeds.

Speed of wind (km/h)	Type of wind
30–49	fresh breeze
50–59	moderate gale
60–69	fresh gale
70–84	strong gale

What type of wind was blowing at 2 am?

- (A) fresh breeze
(B) moderate gale
(C) fresh gale
(D) strong gale

12. Amanda, Bobby, Kabilan and Josh were trying to find out which of the three gymnastics bars in the playground was the hottest. This is what they said.

Amanda: One bar feels the hottest to me, so I already know the answer.

Bobby: All the bars feel hot to me so there cannot be an answer.

Kabilan: It would help if we had a way to measure the temperature of the bars.

Josh: We can take a vote and each of us will vote for the bar they think is the hottest.

Which student gave the best **start** toward finding the answer?

- (A) Amanda
- (B) Bobby
- (C) Kabilan
- (D) Josh

13. A man wanted to work out how tall he was by using the photograph that he took of himself standing next to a scale.



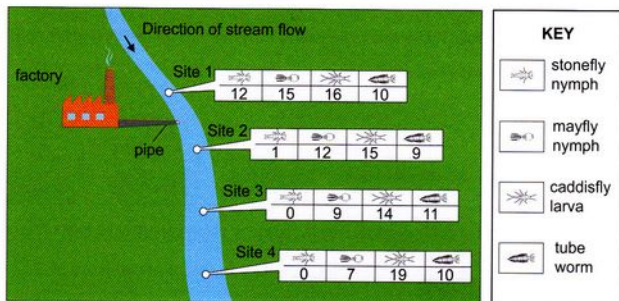
Using the scale, how tall is the man?

- (A) 190 cm
 - (B) 185 cm
 - (C) 170 cm
 - (D) 165 cm
14. Dianne can smell, feel and hear an object inside a bag, but she cannot see it.
- Which of the following is **not** an observation about the object?

- (A) It may taste sweet.
- (B) It smells like banana.
- (C) It has the shape of a cylinder.
- (D) It is rough at one end and smooth at the other.

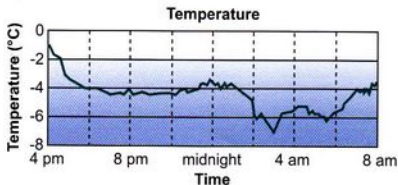
15. Small animals that spend their life under water can be good indicators of pollution. When a stream is polluted, some of these animals will no longer be able to live in the stream.

This diagram shows the number of animals found in a stream at Sites 1 to 4.



Which animal is the best indicator that the stream is polluted?

- (A) tube worm
(B) mayfly nymph
(C) caddisfly larva
(D) stonefly nymph
16. The temperature in an alpine village in Australia was recorded over a number of hours.



What was the lowest temperature, and when was it recorded?

	Lowest temperature ($^{\circ}\text{C}$)	When recorded
(A)	-1	4 pm
(B)	-4	8 pm
(C)	-7	3 am
(D)	-8	midnight

17. This photograph was taken while facing south on Australia's west coast.

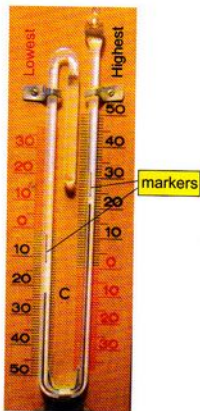


When was the photograph taken?

- (A) in the morning
- (B) about midday
- (C) about midafternoon
- (D) very late in the afternoon

18. A special type of thermometer shows the lowest temperature and the highest temperature reached since it was last reset. The thermometer does this by moving the markers upwards. Then the markers stay in position until they are reset.

The thermometer is reset by moving the markers back down until they both are touching the silver mercury in the U-tube again.



What were the lowest and highest temperatures reached since this thermometer was last reset?

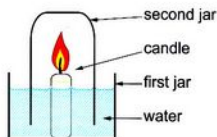
	Lowest temperature (°C)	Highest temperature (°C)
(A)	12	24
(B)	12	22
(C)	18	22
(D)	18	36

19. John studied the temperature over three days.

Time	Temperature (°C)		
	Monday	Tuesday	Wednesday
6 am	6	8	7
8 am	8	9	7
10 am	11	9	8
12 noon	12	11	12
2 pm	15	13	14
4 pm	14	11	13
6 pm	16	10	13
8 pm	12	9	13

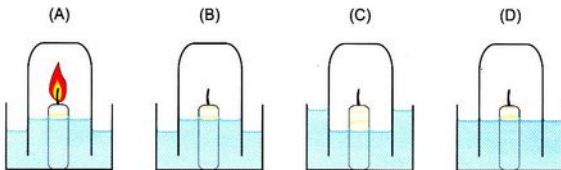
What was the average temperature at 6 pm over the three days?

20. Sunil placed a candle in a jar of water. He then lit the candle and held a second jar over the top.



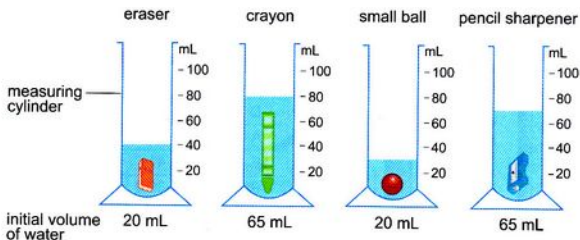
A short time later, the candle went out and water rose into the second jar.

Which diagram shows what happened?



21. A student wanted to see which object had the greatest volume: an eraser, a crayon, a small ball or a pencil sharpener.

Initially a volume of water was placed in four identical empty measuring cylinders. An object was then carefully lowered into the water. The results are shown below.



Which object has the greatest volume?

- (A) eraser
 - (B) crayon
 - (C) small ball
 - (D) pencil sharpener
22. Casey collected five rocks and used a simple method to see if they were magnetic.

She tied string around each rock. She held one end of the string and held the rock over a bench. She placed some objects on the bench below the rock.



Casey predicted that some rocks are more magnetic than others.

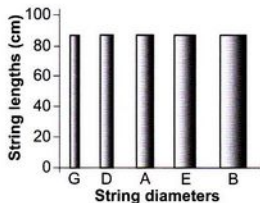
Which of these objects would Casey use to find the most magnetic rock?

- (A) only 1 paper clip
- (B) only 1 wooden craft stick
- (C) as many paper clips as possible
- (D) as many wooden craft sticks as possible

23. A guitar produces sound when its strings vibrate.

The number of times a string vibrates in one second is called its frequency. Frequency is measured in hertz (Hz). The greater the frequency, the higher the pitch of the sound produced.

The graph and table give information about the sound produced by a 5-string bass guitar.

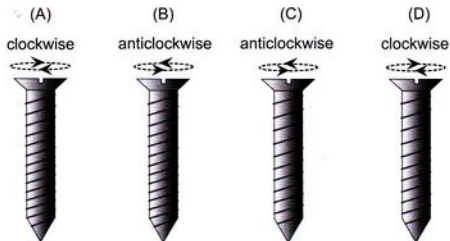


String	Frequency (Hz)
G	98
D	73.5
A	55
E	41
B	30.9

What can be inferred from this information?

- (A) Shorter strings produce lower pitched sound.
(B) Shorter strings produce higher pitched sound.
(C) Thicker strings produce lower pitched sound.
(D) Thicker strings produce higher pitched sound.
24. Mike placed the point of four screws on a piece of wood and turned each screw with a screwdriver in the directions shown.

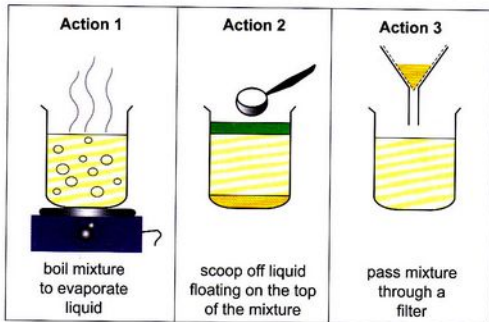
Which screw would go all the way into the wood with the least number of turns?



25. Here is some information about oil, water, sugar and sand.

- Oil floats on water.
- Sugar dissolves in water.
- Sand does not dissolve in water.

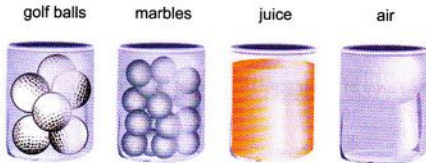
The diagrams show three actions taken to separate a mixture of oil, water, sugar and sand. The actions are not shown in the order they were taken.



In which order were these actions performed to separate each part of the mixture from the others?

- (A) 2, 1, 3
(B) 2, 3, 1
(C) 3, 1, 2
(D) 3, 2, 1

26. Four identical glasses contained the following objects.

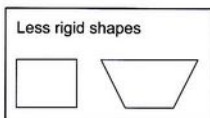
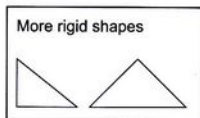


A jug of water is poured into each glass at the same time. The water is poured at the same rate.

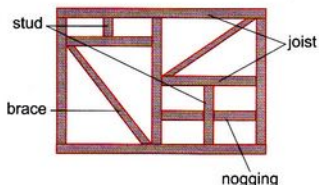
In what order, from first to last, will the glasses overflow?

- (A) air, juice, marbles, golf balls
(B) marbles, juice, air, golf balls
(C) juice, golf balls, marbles, air
(D) juice, marbles, golf balls, air

27. Something that is rigid is hard to push out of shape. Some shapes are more rigid than others.



Bob built a wooden frame from straight pieces of wood as shown.



Which piece of wood was added to the frame to make it more rigid?

28. Joyce poured the same amount of water into four different cylindrical jars.

She measured the surface area and the height of the water in each jar.

The table shows her results.

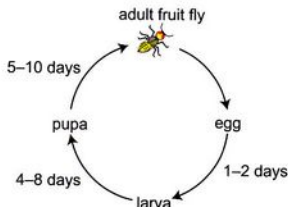
Jar	Surface area of water (cm^2)	Height of water (cm)
1	10	30
2	20	15
3	25	?
4	100	3

What is the height of the water in Jar 3?

- (A) 12 cm (B) 9 cm (C) 6 cm (D) 5 cm

29. The life cycle of an insect is the process of development from when an egg is laid to the time the insect is able to reproduce.

The diagram shows the life cycle of a fruit fly, and gives some facts about fruit flies.



adult fruit flies live 8–10 days and produce 400–500 eggs

fruit flies lay eggs as soon as they become adults

eggs are inserted into the flesh of ripe fruit by female fruit flies

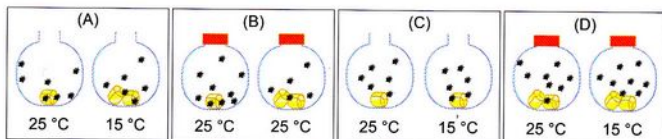
larvae eat fruit

larvae leave fruit in order to pupate

adult fruit flies emerge from the pupae

Simone read that temperature affected the length of the life cycle of fruit flies. She decided to investigate this idea. She trapped some fruit flies and placed them, together with pieces of banana, in glass jars. She observed the fruit flies for two weeks.

Which setup would give Simone a fair test of the effect of temperature on the life cycle of a fruit fly?



30. The diagram shows the weight of four metal objects (Q, R, S and T) being compared using a balance. Each object has the same volume but a different weight.

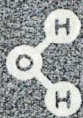


Which further comparison (if any) is required to list these objects in order according to their weight?

- (A) No further comparison is required. (B) S and T
(C) R and T (D) Q and S

SCIENCE 2010 B – ANSWER KEY

Question Number	Paper B
1	C
2	A
3	D
4	C
5	B
6	D
7	D
8	D
9	B
10	B
11	C
12	C
13	C
14	A
15	D
16	C
17	A
18	A
19	C
20	B
21	A
22	C
23	C
24	D
25	B
26	D
27	A
28	A
29	D
30	B



S

2010 Science Answer Keys

ICAS
International Competitions
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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	C	C	A	A	D	C	B	D	A
2	B	A	C	C	C	C	B	A	C
3	B	D	D	A	C	C	A	A	D
4	C	C	D	C	B	B	D	D	A
5	C	B	D	C	B	A	B	C	C
6	B	D	A	D	B	B	A	D	A
7	B	D	B	D	C	A	D	B	C
8	A	D	D	C	B	C	C	D	D
9	D	B	C	B	B	C	C	B	D
10	C	B	B	C	D	D	B	C	B
11	A	C	C	B	C	C	C	A	D
12	A	C	A	D	C	C	C	C	C
13	B	C	C	B	D	D	A	C	D
14	D	A	B	A	A	B	C	B	B
15	D	D	B	A	C	C	B	C	C
16	D	C	B	B	D	A	D	B	C
17	A	A	A	B	A	B	D	D	C
18	C	A	B	D	C	D	A	D	A
19	A	C	C	B	C	B	B	C	C
20	B	B	A	A	A	B	A	A	C
21	D	A	A	B	D	D	A	A	B
22	B	C	C	B	A	A	A	A	C
23	D	C	A	D	A	D	C	B	A
24	A	D	B	A	B	B	D	A	D
25	A	B	B	D	C	A	C	D	A
26	D	D	A	C	C	B	B	A	A
27	C	A	C	C	A	C	C	A	B
28	A	A	C	D	C	B	B	B	B
29	D	D	B	C	B	C	B	D	D
30	C	B	C	B	D	A	B	A	B

(Please turn over)